WEEK 4 CAPSTONE PROJECT

Comprehensive SOC Incident Response Using Wazuh SIEM

Executive Summary

This report demonstrates a full SOC Capstone Project using Wazuh SIEM for threat monitoring, attack detection, investigation, and containment in a virtual lab environment. The project includes attack simulation, log monitoring, failed login detection, threat hunting, and firewall-based containment using practical cybersecurity tools.

Submitted By: Vinayaka Reddy
Repository: cyart-soc-team
Submission Folder: Week4

---

1. Introduction

This project demonstrates a complete Security Operations Center (SOC) workflow in a practical virtual lab environment. Multiple systems were deployed to simulate attacker activity, endpoint monitoring, centralized logging, threat detection, investigation, and incident containment.

The project uses Wazuh SIEM installed on Ubuntu Server, a Windows 11 endpoint with Wazuh agent, Kali Linux as the attacker machine, and Metasploitable2 as the vulnerable target machine.

The purpose of this project is to validate how security teams detect suspicious activity, analyze alerts, and respond quickly to cyber threats.

---

2. Objectives

- Build a practical SOC monitoring lab.
- Simulate reconnaissance and exploitation attempts.
- Detect failed login events using Wazuh SIEM.
- Perform threat hunting and alert investigation.
- Apply containment by blocking attacker IP.
- Conduct root cause analysis.
- Prepare professional incident documentation.

3. Lab Environment

System	Role
Ubuntu Server	Wazuh Manager / SIEM
Windows 11	Endpoint Agent
Kali Linux	Attacker Machine
Metasploitable2	Vulnerable Target

4. Tools Used

- **Wazuh SIEM** – Centralized log monitoring and alerts
- **Kali Linux** – Security testing platform
- **Metasploit Framework** – Exploit simulation
- **Nmap** – Network reconnaissance
- **Ubuntu UFW** – Firewall containment
- **Windows Security Logs** – Endpoint event generation
- **GitHub** – Project submission repository

5. Project Workflow

1. Configure Wazuh Manager on Ubuntu.
2. Connect Windows endpoint to Wazuh agent.
3. Launch attacker machine (Kali Linux).
4. Scan target system using Nmap.
5. Simulate attack activity using Metasploit.
6. Generate Windows failed login attempts (Event ID 4625).
7. Investigate logs in Wazuh SIEM.
8. Block attacker IP using firewall.
9. Perform root cause analysis.
10. Prepare final report.

6. Reconnaissance Activity

The vulnerable Metasploitable2 machine was scanned using Nmap service detection.

Command Used:

nmap -sV 192.168.111.129

The scan identified multiple exposed services including FTP, SSH, Telnet, HTTP, Samba, MySQL, and PostgreSQL.

Figure 1: Nmap Service Enumeration of Vulnerable Target

```
┌──(kali㉿kali)-[~]
└─$ nmap -sV 192.168.111.129
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-27 04:42 -0400
Nmap scan report for 192.168.111.129
Host is up (0.0024s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE     VERSION
21/tcp   open  ftp         vsftpd 2.3.4
22/tcp   open  ssh         OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp   open  telnet      Linux telnetd
25/tcp   open  smtp        Postfix smtpd
53/tcp   open  domain      ISC BIND 9.4.2
80/tcp   open  http        Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp  open  rpcbind
139/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp  open  exec        netkit-rsh rexecd
513/tcp  open  login?
514/tcp  open  shell       Netkit rshd
1099/tcp open  java-rmi    GNU Classpath grmiregistry
1524/tcp open  bindshell   Metasploitable root shell
2049/tcp open  rpcbind
2121/tcp open  ftp         ProFTPD 1.3.1
3306/tcp open  mysql       MySQL 5.0.51a-3ubuntu5
5432/tcp open  postgresql  PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp open  vnc         VNC (protocol 3.3)
6000/tcp open  X11         (access denied)
6667/tcp open  irc         UnrealIRCd
8009/tcp open  ajp13       Apache Jserv (Protocol v1.3)
8180/tcp open  unknown
MAC Address: 00:0C:29:16:0B:DF (VMware)
Service Info: Hosts:  metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_ke

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 31.28 seconds

┌──(kali㉿kali)-[~]
└─$
```

7. Attack Simulation

Metasploit Framework was launched from Kali Linux to identify possible exploit modules and simulate attacker behavior against the vulnerable environment.

Example Commands Used:

msfconsole

search samba

An exploit attempt caused service disruption on the target, demonstrating real risk exposure of vulnerable systems.

Figure 2: Metasploit Framework Exploit Simulation

```
┌──(kali㉿kali)-[~]
└─$ msfconsole
search samba
Metasploit tip: Run modules in the background with run -j so you can
keep working

 =[ metasploit v6.4.116-dev                                 ]
+ -- --=[ 2,623 exploits - 1,326 auxiliary - 1,710 payloads     ]
+ -- --=[ 432 post - 49 encoders - 14 nops - 10 evasion         ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf >
```

8. Detection and Monitoring

The Windows endpoint was connected successfully as an active Wazuh agent. Failed login attempts were intentionally generated to validate detection capability.

Detected Event:

Event ID 4625 – An account failed to log on

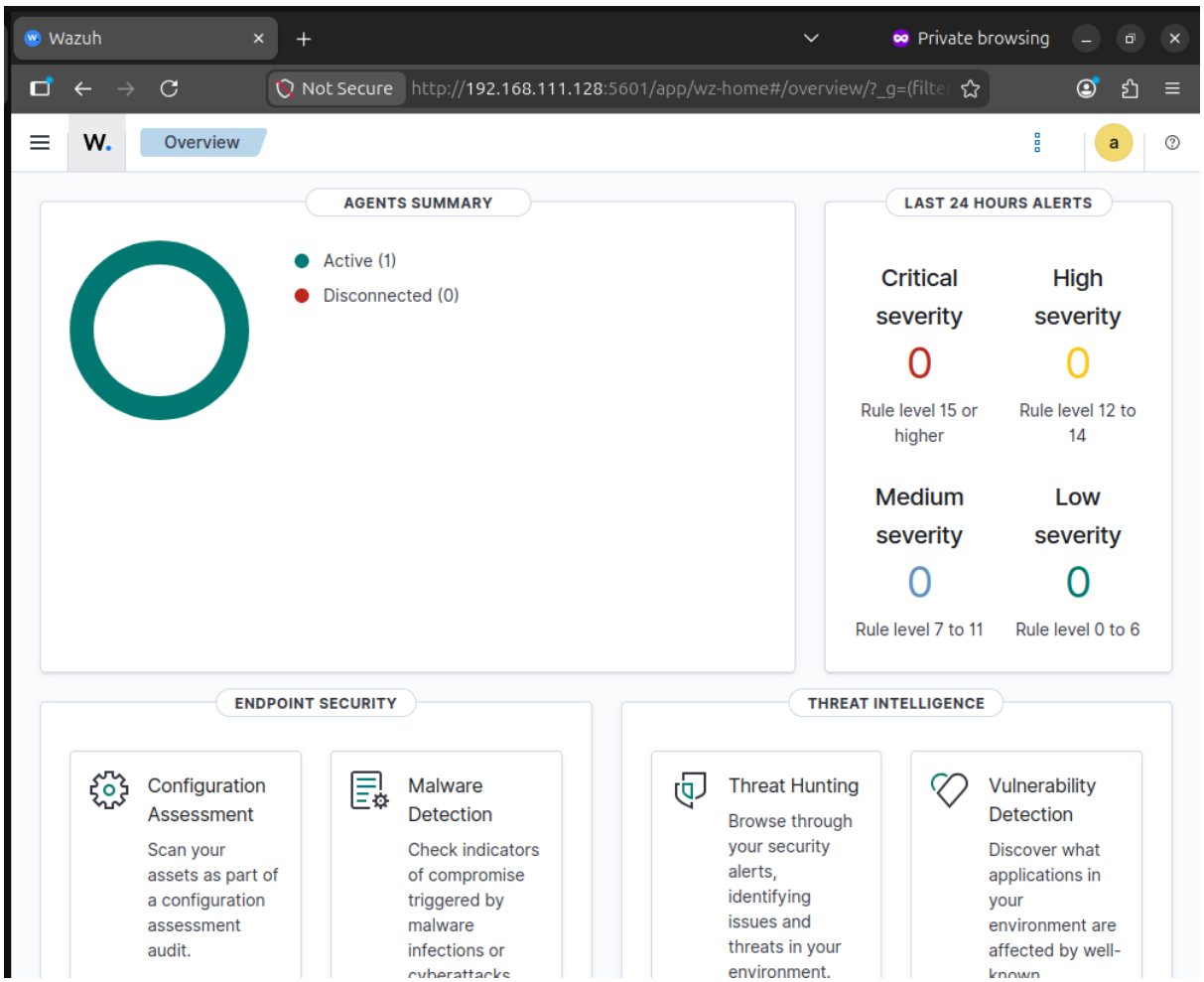
This event is commonly associated with brute-force attempts or invalid credential usage.

Figure 3: Windows Failed Login Event (4625)

Administrator: Command Prompt

```
Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>wevtutil qe Security /q:"*[System[(EventID=4625)]]" /c:5 /rd:true /f:text
Event[0]
  Log Name: Security
  Source: Microsoft-Windows-Security-Auditing
  Date: 2026-04-27T12:54:36.3770000Z
  Event ID: 4625
  Task: Logon
  Level: Information
  Opcode: Info
  Keyword: Audit Failure,
  User: N/A
  User Name: N/A
  Computer: REDDY-PC
  Description:
An account failed to log on.

Subject:
        Security ID:            S-1-5-18
        Account Name:           REDDY-PC$
        Account Domain:         WORKGROUP
        Logon ID:               0x3E7

Logon Type:                     2

Account For Which Logon Failed:
        Security ID:            S-1-0-0
        Account Name:           LapDesk
        Account Domain:         REDDY-PC

Failure Information:
        Failure Reason:         Unknown user name or bad password.
        Status:                 0xC000006D
        Sub Status:             0xC000006A

Process Information:
        Caller Process ID:      0x8b0
        Caller Process Name:    C:\Windows\System32\svchost.exe

Network Information:
        Workstation Name:       REDDY-PC
        Source Network Address: 127.0.0.1
        Source Port:            0

Detailed Authentication Information:
        Logon Process:          User32
        Authentication Package: Negotiate
        Transited Services:     -
        Package Name (NTLM only):       -
        Key Length:             0
```

Figure 4: Wazuh Dashboard Showing Active Endpoint



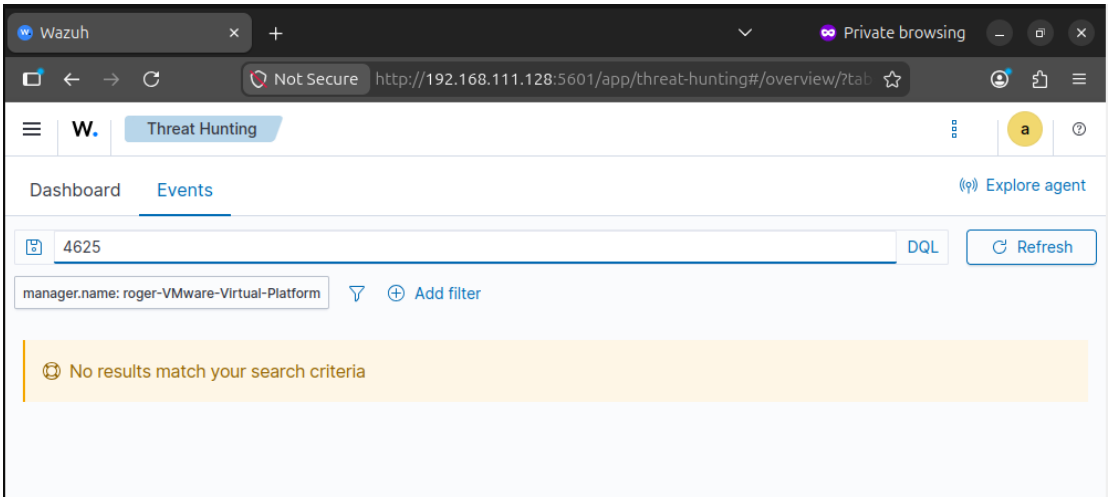
9. Threat Hunting and Investigation

Threat hunting was performed inside Wazuh by reviewing authentication failures and suspicious system behavior.

Focus areas included:

- Repeated failed login attempts
- Source host correlation
- User account anomalies
- Authentication trends
- Suspicious activity timeline

Figure 5: Threat Hunting Search in Wazuh



10. Incident Response and Containment

After suspicious activity was confirmed, the attacker IP address was blocked using Ubuntu UFW firewall.

Command Used:

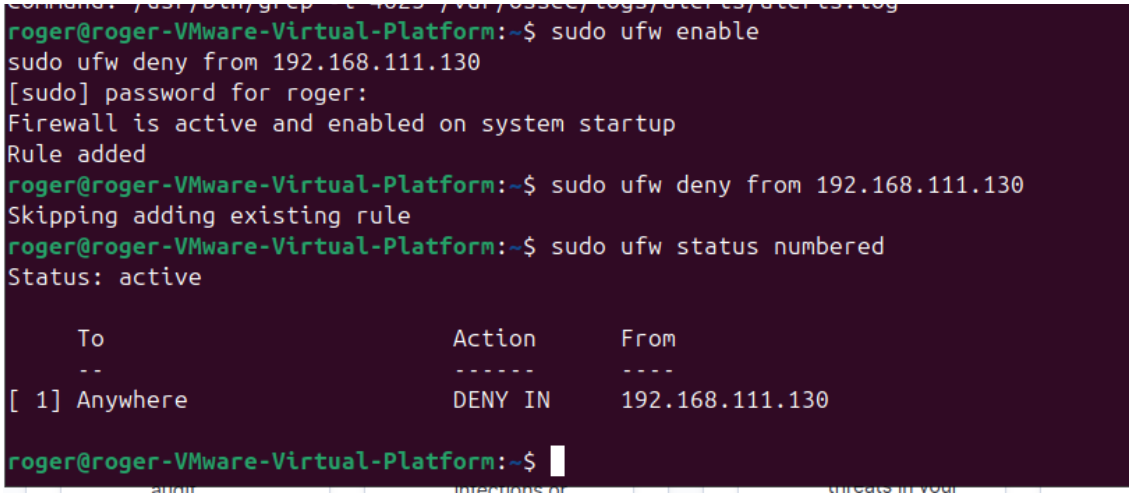
sudo ufw deny from 192.168.111.130

Verification Result:

DENY IN from 192.168.111.130

This containment step prevents further communication from the attacker system.

Figure 6: Firewall Rule Blocking Attacker IP



```
roger@roger-VMware-Virtual-Platform:~$ sudo ufw enable
sudo ufw deny from 192.168.111.130
[sudo] password for roger:
Firewall is active and enabled on system startup
Rule added
roger@roger-VMware-Virtual-Platform:~$ sudo ufw deny from 192.168.111.130
Skipping adding existing rule
roger@roger-VMware-Virtual-Platform:~$ sudo ufw status numbered
Status: active

     To                         Action      From
     --                         ------      ----
[ 1] Anywhere                   DENY IN     192.168.111.130

roger@roger-VMware-Virtual-Platform:~$
```

11.Root Cause Analysis (5 Whys)

Question	Finding
Why did the alert trigger?	Multiple failed logins occurred
Why did failures happen?	Incorrect credentials entered
Why was protection weak?	No account lockout policy
Why was detection important?	Prevent unauthorized access
Why improve controls?	Reduce brute-force risk

12.Security Metrics

Metric	Result
MTTD (Mean Time to Detect)	2 Minutes
MTTR (Mean Time to Respond)	10 Minutes
Severity	High
Final Status	Contained

Severity Justification
High severity was assigned because multiple failed login attempts were detected, indicating possible brute-force or unauthorized access attempts against the monitored endpoint.

13. Recommendations

- Enable account lockout policies.
- Enforce strong passwords.
- Disable unnecessary services.
- Segment lab and production networks.
- Tune SIEM alert rules regularly.
- Perform scheduled vulnerability scans.
- Continue continuous monitoring.

14. Conclusion

This project successfully demonstrated a complete Security Operations Center (SOC) workflow including attack simulation, continuous monitoring, threat detection, investigation, and incident containment. By using Wazuh SIEM, logs from multiple systems were centralized and analyzed effectively, providing improved visibility into security events and suspicious activities.

The practical lab environment helped simulate real-world scenarios such as reconnaissance, exploitation attempts, failed login detections, and firewall-based containment actions. These exercises provided valuable hands-on experience in alert analysis, threat hunting, incident response procedures, and security operations management.

The project also highlighted the importance of proactive monitoring, quick response time, proper log management, and layered security controls in defending organizational systems. Overall, this capstone project enhanced practical cybersecurity skills and demonstrated how SIEM solutions play a critical role in modern SOC environments.

15. References

- Wazuh Official Documentation
- Metasploit Unleashed
- Nmap Documentation
- Ubuntu UFW Firewall Guide
- Microsoft Windows Event ID Reference